

Revised Web Application Project Proposal

1. Project Title

VendorHive: Independent Small Business Co-Op Directory & Cross-Promotion Platform

2. Problem Statement

Local small businesses struggle to compete with e-commerce platforms and large retail chains due to limited digital resources.

- **Local Businesses:** Lack affordable tools for collaboration, joint promotions, and targeted customer reach.
- **Consumers:** Cannot easily discover nearby businesses, receive personalized deals, or get updates from saved businesses.
- **Existing Platforms:** Provide listings, maps, and reviews but do not support cross-promotions, personalized recommendations, or promotion alerts.
- **Customer Feedback:** Reviews exist, but there is no effective way to highlight helpful reviews or integrate feedback with promotions.

Solution:

VendorHive is a unified platform that enables local business discovery, personalized deal recommendations, saved deals, promotion alerts, review management, and merchant-to-merchant cross-promotional campaigns.

3. Proposed Solution

VendorHive is a full-stack MERN application designed to help small businesses aggregate resources, cross-promote, and launch hyper-local marketing campaigns.

- **Core Idea:** A unified platform integrating an interactive public discovery portal for consumers with a merchant management portal for joint promotional campaigns.
- **Target Users:** Independent small business owners (merchants) and neighborhood consumers.
- **System Value:**

- **For Merchants:** Lowers customer acquisition costs through joint promotional agreements with complementary nearby businesses.
- **For Consumers:** Provides a single platform to discover verified local shops, access bundled cross-vendor discounts, and receive tailored deal recommendations.

4. Features & Functional Requirements

A. Authentication & Access Control

- **JWT Security:** Auth via HTTP-only JWTs and bcryptjs password hashing.
- **Role-Based Access:** Permission tiers for Public Users, Merchants, and Admins.

B. Search, Map & Discovery UI

- **Multi-Criteria Filtering:** Filter vendors by category, distance, hours, and active deals.
- **Interactive Mapping:** Mapbox/Leaflet UI rendering pin locations and deals dynamically.
- **Responsive Layout:** Mobile-first React interface with real-time state updates.

C. Personalised Deal Recommendations

- **Location-Based Prioritisation:** Ranks active promotions based on the consumer's real-time or manually selected geographic coordinates.
- **Behavioural Profiling:** Personalises feed based on user bookmark history, preferred categories, viewed deals, and past interaction logs.
- **Recommendation Scoring Engine:** Server-side ranking algorithm that computes a dynamic score based on proximity, category affinity, and overall deal popularity metrics.

D. Saved Deals & Promotion Alerts

- **Bookmarks & Favourites:** Allows consumers to save individual promotional deals and follow preferred merchant profiles.
- **Automated Promotion Alerts:** Automated notification engine sending updates to consumers whenever a followed merchant creates a new promotion.

E. Reviews & Community Feedback

- **Business Reviews & Ratings:** Verified consumers can submit star ratings and written reviews for visited merchants.
- **Review Helpfulness Rating:** Peer-review mechanism allowing users to upvote or downvote reviews as "Helpful" or "Unhelpful" to highlight high-quality feedback.

- **Review Moderation:** Automated keyword filtering alongside merchant and admin flagging pipelines to mitigate spam or inappropriate content.

F. Merchant Dashboard & Management

- **Profile Management:** Interface for merchants to manage business hours, operational contact details, and gallery imagery.
- **Self-Categorization Engine:** Structured taxological tagging system enabling merchants to classify their business (e.g., Restaurant, Clothing, Electronics, Services) for targeted filtering.
- **Cross-Promotion Creator:** Step-by-step form engine to draft and execute joint-vendor promotions (e.g., *"Spend \$20 at Merchant A, get 15% off at Merchant B"*).
- **Deal Management:** Asynchronous CRUD workflow to manage, extend, edit, or deactivate active promotions.

G. RESTful API & Backend Architecture (Node.js / Express.js)

- **Async Processing:** Non-blocking asynchronous handlers using ES6 `async/await` for database operations.
- **Query Engine:** Dedicated routes handling multi-vendor cross-promotion validation, geographical calculations, and recommendation scoring algorithms.
- **Middleware Stack:** Request validation via `express-validator`, custom authorization guards, rate-limiting, and centralized error-handling middleware.
- **Notification Engine:** Background job queue or event-driven service responsible for dispatching promotion alerts to subscribed users.

H. Database Architecture (MongoDB / Mongoose)

- **Data Schemas:** Standardized, strictly typed Mongoose schemas for `Users`, `Merchants`, `Deals`, `Reviews`, and `Notifications`.
- **Geospatial Indexing:** GeoJSON `Point` implementation with MongoDB `$near` and `$geoWithin` operators for real-time spatial calculations.

I. Administration & Platform Metrics

- **Merchant Verification System:** Administrative portal to review, approve, or reject business credentials before allowing merchants to launch deals.
- **Platform Analytics Dashboard:** Visual metrics suite tracking platform activity, total active merchants, active cross-promotions, user engagement, and review trends.

